

40-Day ML · Deep Learning · Computer Vision Internship

Day	Phase	Week	Daily Goal	Topics	Libraries / Tools	Dataset / Platform	Hands-on Tasks
PHASE 1 — MACHINE LEARNING FOUNDATIONS (Days 1–10)							
1	Phase 1	Wk 1	Set up ML environment and perform full EDA on a real dataset	Python setup, Jupyter, NumPy, Pandas, Matplotlib, Seaborn — distributions, correlations, missing-value heatmaps	NumPy 1.26, Pandas 2.x, Matplotlib, Seaborn, Jupyter Lab	Kaggle – Titanic (tabular CSV)	Install env (conda/venv); load Titanic CSV; plot age/fare distributions; correlation heatmap; identify missing columns; write 5-line findings summary
2	Phase 1	Wk 1	Build a preprocessing pipeline with zero data leakage	ColumnTransformer, SimpleImputer, OneHotEncoder, StandardScaler, Pipeline, train_test_split with stratify	scikit-learn 1.4 (Pipeline, ColumnTransformer, SimpleImputer)	Kaggle – House Prices (tabular CSV)	Identify numeric vs categorical cols; build ColumnTransformer; wrap in Pipeline; confirm no leakage by fitting only on train; pickle the pipeline
3	Phase 1	Wk 1	Train and evaluate Linear & Logistic Regression; interpret coefficients	Linear Regression (OLS), Lasso, Ridge, Logistic Regression, confusion matrix, classification report, RMSE, R ²	scikit-learn (LinearRegression, LogisticRegression, Ridge, Lasso), statsmodels	Kaggle – House Prices + Titanic	Train LR on House Prices; print RMSE/R ² ; train LogReg on Titanic; print F1/AUC; plot coefficient bar chart; interpret top-3 features
4	Phase 1	Wk 1	Understand and tune tree-based models; visualise feature importance	DecisionTreeClassifier (max_depth, min_samples_leaf), RandomForestClassifier, feature_importances_, overfitting vs underfitting	scikit-learn (DecisionTreeClassifier, RandomForestClassifier), dtreeviz, matplotlib	Kaggle – Titanic	Train DT at depth 3 vs 10; plot decision tree; train RF with 100 trees; print top-10 feature importances; compare train vs val accuracy

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5	Phase 1	Wk 1	Master SVM kernels and build a text Naive Bayes classifier	SVC (linear, RBF, poly), kernel trick intuition, GaussianNB, MultinomialNB, TfidfVectorizer, precision-recall	scikit-learn (SVC, MultinomialNB, TfidfVectorizer), matplotlib	UCI SMS Spam Collection (Kaggle or UCI ML Repo)	Train SVM linear vs RBF on Iris; plot decision boundary; build TF-IDF + MultinomialNB spam classifier; report precision/recall/F1
6	Phase 1	Wk 2	Apply PCA for dimensionality reduction and cluster customers with K-Means	PCA (explained variance, scree plot), KMeans (elbow method, silhouette score), DBSCAN, t-SNE visualisation	scikit-learn (PCA, KMeans, DBSCAN, TSNE), yellowbrick, seaborn	Kaggle – Mall Customer Segmentation	Run PCA; plot cumulative explained variance; choose n_components; run KMeans k=2–8; plot elbow + silhouette; visualise clusters with t-SNE
7	Phase 1	Wk 2	Implement rigorous model evaluation: cross-validation, ROC, calibration	StratifiedKfold, cross_val_score, roc_auc_score, precision_recall_curve, calibration_curve, classification_report	scikit-learn (StratifiedKfold, roc_curve, calibration), matplotlib	Kaggle – Credit Card Fraud Detection	Run 5-fold CV on 3 classifiers; plot ROC curves on same axes; plot PR curve (imbalanced data); report mean ± std AUC
8	Phase 1	Wk 2	Engineer meaningful features and select the most informative subset	Polynomial features, log-transform skewed cols, date decomposition, SelectKBest (chi2, f_classif), RFECV	scikit-learn (PolynomialFeatures, SelectKBest, RFECV), feature-engine, pandas	Kaggle – Titanic or House Prices	Create 5 new features (title from name, fare_per_person, age_band etc.); run SelectKBest; run RFECV; compare baseline accuracy vs engineered accuracy

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9	Phase 1	Wk 2	Tune hyperparameters efficiently with Optuna and validate best config	GridSearchCV, RandomizedSearchCV, Optuna (TPE sampler), learning curves, validation curves	scikit-learn, Optuna 3.x, joblib (n_jobs=-1)	Kaggle – House Prices	GridSearch RF (5 params); Optuna 50 trials RF; compare results + time cost; plot Optuna optimisation history; log best params to JSON
10	Phase 1	Wk 2	Submit a complete end-to-end ML solution to a Kaggle competition	Full pipeline: EDA → feature engineering → model selection → tuning → submission CSV	scikit-learn, XGBoost, LightGBM, pandas, Optuna	Kaggle – Titanic or Spaceship Titanic competition	Train XGBoost + LightGBM; ensemble with VotingClassifier; generate submission.csv; submit to Kaggle; screenshot leaderboard score
PHASE 2 — DEEP LEARNING (Days 11–22)							
11	Phase 2	Wk 3	Implement a neural network from scratch to understand backpropagation	Perceptron, sigmoid/ReLU, forward pass, loss (MSE/cross-entropy), gradient descent, chain rule, weight update	NumPy only (no frameworks), matplotlib for loss curves	sklearn datasets – make_moons or MNIST (load locally)	Build 2-layer NN class in NumPy; implement forward + backward pass; train on make_moons; plot decision boundary after each 100 epochs
12	Phase 2	Wk 3	Build and train an MLP in Keras exceeding 97% accuracy on MNIST	Sequential API, Dense, BatchNormalization, Dropout, EarlyStopping, ModelCheckpoint, sparse_categorical_crossentropy	TensorFlow 2.x / Keras, TensorBoard	TensorFlow Datasets (TFDS) – MNIST	Build 3-layer MLP; add BN + Dropout; use EarlyStopping; log to TensorBoard; achieve >97% test accuracy; save model as .keras file

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13	Phase 2	Wk 3	Diagnose training dynamics and tune learning rate + optimizer	LR finder, ReduceLROnPlateau, CosineDecay, Adam/SGD/RMSprop comparison, loss landscape intuition	TensorFlow/Keras (callbacks), keras-tuner, matplotlib	TFDS – CIFAR-10	Run LR range test; plot loss vs LR; compare Adam/SGD on same model; add ReduceLROnPlateau; log val_loss improvement in a table
14	Phase 2	Wk 3	Prevent overfitting with Dropout, Batch Norm, L2 and data augmentation	Dropout, BatchNormalization, L1/L2 kernel regularizer, train vs val gap analysis, augmentation with tf.image	TensorFlow/Keras, matplotlib	TFDS – CIFAR-10	Train baseline → add Dropout → add BN → add L2; plot 4 val-accuracy curves on one graph; identify best combo
15	Phase 2	Wk 3	Build a CNN from scratch and visualise learned filters + feature maps	Conv2D, MaxPooling2D, GlobalAveragePooling2D, padding, stride, receptive field, filter visualisation	TensorFlow/Keras, matplotlib, keras-viz (grad-cam)	TFDS – CIFAR-10	Build 5-layer CNN; train 20 epochs; visualise 32 filters of Conv1; plot feature maps for 3 test images; compare CNN vs MLP accuracy
16	Phase 2	Wk 4	Fine-tune MobileNetV2 on a custom dataset with frozen + unfrozen layers	Transfer learning, feature extraction vs fine-tuning, base_model.trainable, layer freezing strategy	TensorFlow/Keras (MobileNetV2, EfficientNetB0), keras preprocessing	Kaggle – Flowers Recognition (5 classes) or any custom image folder	Load MobileNetV2 imagenet weights; freeze base; train top layers 10 epochs; unfreeze last 30 layers; fine-tune 10 more epochs; compare accuracy

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17	Phase 2	Wk 4	Build a robust image augmentation pipeline and measure its impact	RandomFlip, RandomRotation, RandomZoom, RandomContrast, Albumentations (ShiftScaleRotate, HueSaturationValue)	TensorFlow/Keras (tf.keras.layers augmentation), Albumentations 1.x	Kaggle – Flowers Recognition	Build augmentation layer; visualise 16 augmented images; train model with/without augmentation; plot accuracy comparison; measure +% gain
18	Phase 2	Wk 4	Build an LSTM sentiment classifier and understand sequence dynamics	Embedding layer, LSTM, GRU, Bidirectional, TextVectorization, padding/truncation, BPTT	TensorFlow/Keras (Embedding, LSTM, Bidirectional), NLTK (tokenize)	Kaggle – IMDB Movie Reviews (50k)	Tokenize with TextVectorization; build Embedding+BiLSTM; train 10 epochs; plot accuracy/loss; compare LSTM vs GRU; print 5 misclassified reviews
19	Phase 2	Wk 5	Fine-tune a pretrained Transformer for text classification	Tokenizer, attention mask, BERT/DistilBERT architecture, AutoModelForSequenceClassification, Trainer API	HuggingFace Transformers 4.x, Datasets (HF Hub), PyTorch or TF backend	HuggingFace Hub – imdb or tweet_eval dataset	Load DistilBERT tokenizer + model; tokenize dataset; fine-tune 3 epochs; evaluate accuracy + F1; save model; compare with LSTM baseline
20	Phase 2	Wk 5	Export a trained model to TF-Lite and benchmark inference speed	SavedModel format, TFLiteConverter, quantization (float16, int8), latency benchmarking	TensorFlow Lite, tf.lite.TFLiteConverter, time module	Use model from Day 16 (Flowers MobileNetV2)	Export to SavedModel; convert to TF-Lite (float32 then int8 quant); measure inference time on 100 images; compare accuracy vs full model

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21	Phase 2	Wk 5	Deploy a DL model as a REST API and test with Postman	Flask routing, request/response JSON, base64 image decoding, model loading, error handling, CORS	Flask 3.x, Pillow, NumPy, gunicorn, Postman	Use model from Day 16 or 20	Write Flask app with /predict endpoint; decode base64 image; run inference; return class + confidence JSON; test 5 images via Postman; handle errors
22	Phase 2	Wk 5	Package DL project with GitHub README, demo notebook and requirements	GitHub repo structure, requirements.txt, .gitignore, demo Jupyter notebook, model card	Git, GitHub, Jupyter, pip freeze	All datasets used in Phase 2	Create clean GitHub repo; write README (problem, dataset, results table, usage); add demo notebook; push model weights; write model card
PHASE 3 — COMPUTER VISION (Days 23–34)							
23	Phase 3	Wk 6	Master OpenCV image I/O, colour spaces and geometric transforms	cv2.imread/imshow/imwrite, BGR↔RGB↔HSV↔GRAY, resize/crop/rotate/flip, warpAffine, perspective transform	OpenCV 4.x (cv2), NumPy, matplotlib	Local photos / OpenCV sample images (cv2.data.harcascades)	Load 5 images; convert all colour spaces; apply resize+crop+rotate; apply perspective warp on a book/document photo; save results
24	Phase 3	Wk 6	Apply spatial filters and detect edges with Canny and Sobel operators	GaussianBlur, medianBlur, bilateralFilter, Sobel (dx,dy), Laplacian, Canny (thresholds), Fourier transform	OpenCV 4.x (cv2), NumPy, matplotlib	OpenCV sample images or personal photos	Apply 4 blur types; apply Sobel X/Y/combined; apply Canny at 3 threshold pairs; compare results in a 4×3 matplotlib grid; explain best for each use case

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25	Phase 3	Wk 6	Segment objects with thresholding and morphological operations; count them	Otsu threshold, adaptiveThreshold, erode/dilate/open/close, findContours, drawContours, cv2.moments, object counting	OpenCV 4.x (cv2), NumPy	Google Images – coins on white background (or blood cell dataset on Kaggle)	Load coin image; apply Otsu threshold; erode+dilate; find contours; draw bounding boxes; print object count; compare manual vs automatic count
26	Phase 3	Wk 6	Detect and match keypoint features between two images	ORB, SIFT, FLANN matcher, BFMatcher, Lowe's ratio test, homography (findHomography, RANSAC), image stitching	OpenCV 4.x contrib (cv2.SIFT_create, cv2.ORB_create), NumPy	Personal photos of the same scene from 2 angles	Extract ORB keypoints from 2 photos; match with BFMatcher; apply ratio test; compute homography; warp and stitch images; visualise matches
27	Phase 3	Wk 6	Run YOLOv8 inference and understand output format (boxes, scores, classes)	YOLO architecture (backbone, neck, head), anchor-free detection, NMS, mAP@0.5, confidence threshold, IoU	Ultralytics YOLOv8 (pip install ultralytics), OpenCV, PIL	COCO val images (download 5–10) or personal photos	pip install ultralytics; run yolov8n.pt on 5 images; parse results (boxes, conf, class); draw custom bounding boxes with cv2; print class counts
28	Phase 3	Wk 7	Train YOLOv8 on a custom dataset and reach mAP > 0.60	Dataset labelling (YOLO format), data.yaml config, training epochs, batch size, imgsz, val mAP@0.5, confusion matrix	Ultralytics YOLOv8, Roboflow (labelling + export), Google Colab T4 GPU	Roboflow Universe – any small dataset (100–300 images, 2–5 classes, e.g. helmet or mask detection)	Export dataset from Roboflow in YOLOv8 format; train yolov8s 30 epochs; log mAP curve; test on 5 unseen images; save best.pt

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29	Phase 3	Wk 7	Build a face detection + recognition system with identity lookup	Haar cascade, dlib HOG face detector, face_recognition (128-d embeddings), cosine similarity, identity matching	OpenCV (cv2.CascadeClassifier), face_recognition (dlib), NumPy	LFW dataset (scikit-learn fetch_lfw_people) or personal photo folder	Detect faces in 10 images with Haar; compare dlib accuracy; register 5 identities; run recognition on new photo; print name + confidence
30	Phase 3	Wk 7	Detect body pose landmarks in real-time and compute joint angles	MediaPipe Pose (33 landmarks), landmark normalisation, joint angle calculation (cos rule), FPS counter	MediaPipe 0.10.x (mediapipe), OpenCV, NumPy, math	Own webcam video or YouTube workout clip (downloaded with yt-dlp)	Run mp.solutions.pose on video; draw skeleton; compute elbow angle frame-by-frame; plot angle over time; measure real-time FPS
31	Phase 3	Wk 7	Run semantic segmentation and understand per-pixel class masks	Semantic vs instance segmentation, DeepLabV3+ (ResNet101 backbone), pixel accuracy, mIoU, colormap overlay	TensorFlow Hub (DeepLabV3+ model), OpenCV, NumPy, matplotlib	Cityscapes sample images or ADE20K (5 images from Kaggle)	Load DeepLabV3+ from TF Hub; run on 5 images; colour-map each class; compute pixel accuracy; overlay transparent mask on original image
32	Phase 3	Wk 8	Extract text from documents and images with OCR pipelines	Tesseract (PSM modes, OEM), EasyOCR (multi-language), image pre-processing for OCR (binarise, deskew), bounding boxes	EasyOCR 1.7, pytesseract, OpenCV, Pillow	Own scanned document / ID card mockup or FUNSD dataset (Kaggle)	Pre-process image (grayscale, threshold, skew); run EasyOCR; run Tesseract PSM 6; compare accuracy on 5 docs; draw word-level bboxes

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33	Phase 3	Wk 8	Build a real-time object-counting pipeline at ≥20 FPS	Video capture loop, YOLOv8 inference per frame, per-class counter, FPS overlay, cv2.VideoWriter (output video)	Ultralytics YOLOv8, OpenCV 4.x, collections.Counter	Downloaded street/traffic video (Pexels free videos) or own webcam	Open video; run YOLOv8n per frame; overlay bboxes + class labels + live count; display FPS; save annotated output video; report avg FPS
34	Phase 3	Wk 8	Package and demo the CV mini-project with a recorded walkthrough	Project structure, argparse CLI, requirements.txt, README with demo GIF, Loom/OBS recording	argparse, Loom or OBS Studio, GitHub, ffmpeg (GIF creation)	All Phase 3 data / webcam	Refactor best CV project into clean CLI app; record 2-min demo video; create demo.gif for README; push to GitHub with full README
PHASE 4 — CAPSTONE + PORTFOLIO (Days 35–40)							
35	Phase 4	Wk 9	Define capstone problem, select dataset and set up tracked experiment	Problem framing (ASD detection / CV health system), literature review (2 papers), MLflow experiment setup, GitHub milestones	MLflow 2.x, GitHub Projects, arXiv, Papers With Code	Kaggle / Roboflow / UCI — dataset matching problem (e.g. ASD behavioural dataset)	Write 1-page project brief (problem, dataset, metrics, timeline); create MLflow experiment; create GitHub repo with Issues & milestones; skim 2 papers
36	Phase 4	Wk 9	Build preprocessing pipeline and establish a strong baseline model	Full data pipeline, baseline model (XGBoost or ResNet), MLflow run logging (params, metrics, artifacts)	MLflow, scikit-learn or TF/Keras, pandas, NumPy	Chosen capstone dataset	Clean + split data; train baseline model; log accuracy/F1/loss to MLflow; commit pipeline code; open GitHub PR for baseline branch

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37	Phase 4	Wk 9	Integrate DL and CV components; run ablation experiments	Model architecture choices, ablation study (component on/off), MLflow compare runs, experiment tracking best practices	MLflow (compare_runs), TF/Keras, Ultralytics or OpenCV (CV module)	Capstone dataset + augmented data	Add DL/CV module to pipeline; run 3 ablation configs; log each to MLflow; compare run metrics in MLflow UI; pick best config
38	Phase 4	Wk 9	Evaluate final model thoroughly and write technical documentation	Final metrics (accuracy, F1, AUC, mAP), confusion matrix, error analysis (false positives/negatives), README, model card	matplotlib, seaborn (confusion matrix), MLflow, Jupyter, GitHub	Capstone dataset test split	Generate full evaluation report; analyse 10 failure cases; write README (background, method, results table, limitations); create model card
39	Phase 4	Wk 10	Write a structured research summary and update professional portfolio	Abstract, methodology, results, conclusion (2-page format), LinkedIn project section, GitHub pinned repos	Overleaf (LaTeX) or Google Docs, LinkedIn, GitHub	All project results	Write 2-page technical summary (400 words + figures); update LinkedIn with project + skills; pin 3 repos on GitHub; draft 3-line project description
40	Phase 4	Wk 10	Present capstone with a polished demo and live Q&A	10-slide deck structure, live demo script, anticipate questions (model choice, limitations, future work), delivery practice	Google Slides or Canva, Loom (demo video), GitHub Pages (optional)	All capstone data and models	Build 10-slide deck; record 3-min demo video; rehearse 2× full presentation; present to supervisor; log feedback; plan next steps